# Week 15 — Extended Reading: Sequence and diffusion policies

STUDY-RL

Source audit: 20 October 2026

## Read for a claim, an assumption and a test

Read the teaching guide first, then use this pack alongside the lecture. The annotations below are course-written paraphrases and investigation prompts, not quotations from the papers. Each source link identifies a primary paper or official guidance. Benchmark results do not transfer automatically to our small exercises. For each reading, write the mathematical object, the needed assumptions and one observation that would challenge your interpretation.

## 1 Chen et al. (2021): Decision Transformer

**Primary source.** https://arxiv.org/abs/2106.01345

**Mechanism and scope.** DT models actions from a causally ordered history containing return-to-go, states and past actions. Conditioning selects behaviour represented in the training data; requesting a large return does not ensure that the environment will deliver it.

**Read, derive, investigate.** Write the tokens available before action $a_t$. Which reward is future information during training, and how is the desired target supplied at inference? Audit action-position loss masks so that the action being predicted cannot leak through its own input token.

## 2 Janner et al. (2021): Trajectory Transformer

**Primary source.** https://arxiv.org/abs/2106.02039

**Mechanism and scope.** The Trajectory Transformer models discretised trajectory components and uses search for planning. Unlike DT's direct return-conditioned action prediction, it can score candidate continuations through a learned sequence model.

**Read, derive, investigate.** Distinguish model likelihood from environment reward. Explain how beam width, discretisation and horizon affect search, and why a high-scoring predicted future may exploit model error. The small course DT is not a Trajectory Transformer implementation.

## 3 Janner et al. (2022): Diffuser

**Primary source.** https://arxiv.org/abs/2205.09991

**Mechanism and scope.** Diffuser learns trajectory generation by iterative denoising and uses conditioning or guidance to shape plans. Denoising steps and environment time steps are different axes. Guidance changes sampling and can push a generated plan away from familiar data.

**Read, derive, investigate.** Draw a matrix with denoising iteration horizontally and environment time vertically. Mark which entries are clamped by a known current state. Why does executing only the next action and replanning reduce, but not eliminate, model error? Do not describe iterative denoising as a single forward pass.

## 4 Chi et al. (2023): Diffusion Policy

**Primary source.** https://arxiv.org/abs/2303.04137

**Mechanism and scope.** Diffusion Policy represents conditional action distributions using diffusion and action sequences in robotic imitation learning. Its evaluation concerns specific manipulation tasks and data; multimodal action modelling is not itself reinforcement-learning improvement.

**Read, derive, investigate.** Compare a bimodal action distribution to a squared-error regressor that averages its modes. When could the mean action be poor? Separate action-horizon length, observation conditioning and denoising cost. Our toy density experiment does not reproduce the robot experiments.

## 5 Hansen-Estruch et al. (2023): IDQL

**Primary source.** https://arxiv.org/abs/2304.10573

**Mechanism and scope.** IDQL connects implicit value learning with policy extraction and uses a diffusion behaviour model for expressive action proposals. Its extraction perspective helps explain why the critic and proposal distribution must be considered together.

**Read, derive, investigate.** If every sampled proposal comes from a region with mediocre value, can reranking recover an unproposed action? Trace proposal generation, critic scoring and final selection separately. Compare this to Diffusion-QL, where a Q term affects policy training rather than being only an inference-time guide.

## 6 Brandfonbrener et al. (2022): When does return conditioning work?

**Primary source.** https://arxiv.org/abs/2206.01079

**Mechanism and scope.** This analysis studies conditions behind return-conditioned supervised learning and its failures. A favourable trajectory can reflect environment luck rather than an action that reliably causes a favourable outcome.

**Read, derive, investigate.** Construct two actions with identical observed high-return trajectories but different expected returns. Which conditional-support or near-determinism premise is missing? Use this question to interpret a target-return sweep, rather than assuming achieved return must increase monotonically.

## Connect the reading to your workbook

Recompute an undiscounted return-to-go by backward summation, then change the mathematical convention to $G_t = r_t + \gamma G_{t+1}$. The target update becomes $(G_t - r_t)/\gamma$ for $\gamma > 0$. The shipped helper uses $\gamma = 1$; label any discounted extension explicitly. Compare requested and achieved returns on held-out episodes.

**Evidence boundary.** Source identities and the central claims above were checked against primary material. These annotations do not reproduce every proof, training run or empirical result. Use the original paper for theorem assumptions and experimental details; use `project/` and the workbook for the exact course implementation.